

56-362-250



**Power
Generation**

Cummins India Ltd.

Power Generation Business Unit

Engine Test Certificate

This is to certify that Engine Model **QSB7-G5**

Serial Number **84495547**was tested using

Diesel as per ISO - 3046 Standard.

Corrected Full Load B.H.P to N.T.P conditions as per the

test standard is..... **285** B.H.P. @ **1500** R.P.M.
324BHP @ 1800RPM

Date : **29-Jun-18**
/ /

Nilcei
Product Conformance

A048U279

	Precision Industries Testing Department	GENERATING-SET LOAD TEST				Model No: PI 250 C	60 Hz
					Type: Close	Job No: 9522	
					Color: White	Date: 01/11/2018	

Serial Number:								
PI Model No:	PI 250 C	Prime Power:	228	[kVA]	P.F (Load):	1	Voltage:	220

Generating Set Details:

Engine:	Cummins	Model No: QSB7 - G5	Serial No: 84495547
Alternator:	Stamford	Model No: UCI 274 H1	Serial No: N18G271024
Controller:	DSE	Model No: 6120	Serial No: 6865754
C. Breaker:	Metasol	Model No: ABN 803C	Serial No: N/A
C. Transformer:	Ziegler 800/5A	Model No: N/A	Serial No: N/A
AVR:	Stamford	Model No: SX 460	Serial No: 1121010 - 0803
Cooling System: Water Cool	Enclosure Type: Canopy		Protection Class: IP23

Load Test Ambient Conditions:

Temperature	33	[°C]	Altitude:	0	[m]	Humidity:	40	%
-------------	----	------	-----------	---	-----	-----------	----	---

Generating Set Rating:

Testing Full Load Current:	478.68	Amps	Nominal Current:	598.3	Amps
Power Correction Factor (De-rating):	0.000	%	Corrected Prime Power:	0.0	[kVA]

LOAD TEST

Load	Run Time	Freq	Speed	VOLTAGE L-L			CURRENT			Power	Coolant Temp	Oil Pressure	Charg-Alt.
				L1	L2	L3	L1	L2	L3				
%	min	Hz	RPM	V	V	V	A	A	A	kW	C	bar	VDC
0 %	5	60	1800	220	220	220	-	-	-	-	62	4.85	14.0
25 %	5	60	1800	220	220	220	118	118	118	45.0	68	4.80	14.0
50 %	5	60	1800	220	220	221	240	240	240	91.0	74	4.76	14.0
75 %	5	60	1800	220	220	221	358	358	358	137.0	80	3.52	14.0
100 %	10	60	1800	220	221	221	475	474	475	181.0	84	3.40	14.0
110 %	10	60	1800	220	221	221	525	526	525	201.0	89	3.24	14.0

LOAD ACCEPTANCE TEST

Suddenly Block Load %	Voltage [V]	Frequency [Hz]	Current [A]	Response Time [sec]
50%	220	60.0	240	0
0%	225 - 220	61.0 - 60.0	0	2
50%	212 - 220	57.8 - 60.0	242	2

PROTECTION SYSTEM TEST

Over Frequency Warning / SD	Under Frequency Warning / SD	Over Voltage Warning / SD	Under Voltage Warning / SD	Oil Pressure SD	Coolant Level SD	Coolant Temp Warning / SD
65 / 67	55 / 53	240 / 260	190 / 180	1.0	ECM OK	103 / 108 c



TESTED

Testing Department

ACCESSORIES DETAILS

Model:	Serial No.
Model:	Serial No.

PI Document no.	PI-TST-LT-01	Rev.	01
-----------------	--------------	------	----

Test Certificate

FOR

SALIENT POLE, SELF EXCITED / PMG EXCITED AND SELF REGULATED AC GENERATOR

Machine No. **N18G271024**

Type Test has been conducted on similar design machine
STANDARDS: Generator generally conforms to BS 5000 : Part 3,
 Tested in accordance with BS EN 60034-1/IS/IEC 60034-1
 Rotor dynamically balanced to BS ISO 1940 - 1

ROUTINE TEST CERTIFICATE FOR BRUSHLESS AC GENERATOR:

FRAME: UCI274H1	AVR: SX460	AVR NO: 1121010803	ENCLOSURE: IP23	
RATED KVA: 200	RATED VOLTS: 400	RATED AMPS: 288.7	FREQUENCY: 50	PHASE: 3
POWER FACTOR: .8	STR CONN: Series Star	STR WINDING: 311	DUTY: Base Continuous	WIRE: 4-Wire
EXC. VOLTS: 58	EXC.AMPS: 2.3	AMB TEMP: 40		

FOR 3 PHASE SEQUENCE LOOKING FROM DRIVE END FOR CLOCKWISE ROTATION :U-V-W

WINDING	COLD RESISTANCE IN Ohm(+/- 10 Milli Ohm) AT 20°C (MAX VALUES)	INSULATION RESISTANCE (M- Ohm)	HV TEST 2KV FOR 1 MIN	INSULATION CLASS
STATOR	0.03371	2+	OK	H
ROTOR	1.9663	2+	OK	H

REGULATION TEST

FREQ.	VOLTS	AMPS	KVA	PF	KW	EXC.VOLT	EXC.AMP	% LOAD
52	401.35	0.00	0.00	0.00	0.00	12	.5	0.00
50	400	288.7	200	.8	160	58	2.3	100.00

REGULATION = (N.L.VOLTAGE-F.L.VOLTAGE / RATED VOLTAGE)*100 = 0.34 %

Item Description: AC Generator.UC27.4.H.1.311.,3,,11.5.,SX460.Black - RAL9011
 Remarks:

Date: 06-JUL-18	Job No: 3525767-008	AUTHORISED BY Shrikant B Labade Quality Assurance
-----------------	---------------------	--

Certificate Of Conformity

Quality System Approval
 ISO 9001/2008

Frame Size : **UCI274H1**Serial Number : **N18G271024**

Manufactured Date : 06-JUL-18

KVA: **200**Authorized By Quality Assurance: **Shrikant B Labade**

EU DECLARATION OF CONFORMITY



This synchronous A.C. generator is designed for incorporation into an electricity generating-set and fulfils all the relevant provisions of the following EU Directive(s) when installed in accordance with the installation instructions contained in the product documentation:

2014/35/EU	Low Voltage Directive
2014/30/EU	The Electromagnetic Compatibility (EMC) Directive

and that the standards and/or technical specifications referenced below have been applied:

EN 61000-6-2:2005	Electromagnetic compatibility (EMC). Generic standards - Part 6-2: Immunity for industrial environments
EN 61000-6-4:2007+A1:2011	Electromagnetic compatibility (EMC). Generic standards - Part 6-4: Emission standard for industrial environments
EN ISO 12100:2010	Safety of machinery - General principles for design - Risk assessment and risk reduction
EN 60034-1:2010	Rotating electrical machines - Part 1: Rating and performance
BS ISO 8528-3:2005	Reciprocating internal combustion engine driven alternating current generating sets - Part 3: Alternating current generators for generating sets
BS 5000-3:2006	Rotating electrical machines of particular types or for particular applications - Part 3: Generators to be driven by reciprocating internal combustion engines - Requirements for resistance to vibration

This declaration has been issued under the sole responsibility of the manufacturer. The object of this Declaration is in conformity with the relevant Union harmonization Legislation.

The name and address of authorised representative, authorised to compile the relevant technical documentation, is the Company Secretary, Cummins Generator Technologies Limited, 49/51 Gresham Road, Staines, Middlesex, TW18 2BD, U.K.

Signed:  Date: 01st February 2016	Name, Title and Address: Kevan J Simon Global Technical and Quality Director Cummins Generator Technologies Fountain Court Lynch Wood Peterborough, UK PE2 6FZ
---	--

Description UCI274H1

Serial Number N18G271024

450-16383-F